

site class under π as in its own.⁸ To see that a vertex $v \in G - U$ is happy with π , let i be minimal such that $v \in G_{n_i}$ and recall that v was happy with π_{n_i} . As both v and its neighbours in G lie in $U \cup V(G_{n_i} - G_{n_{i-1}})$, and π agrees with π_{n_i} on this set, v is happy also with π . Vertices in U_0 are happy with π , because they were happy with π_{n_0} , and π agrees with π_{n_0} on U_0 and all its neighbours. Vertices in U_1 are also happy. Indeed, every $u \in U_1$ has infinitely many neighbours in some C_n , and hence in some G_{n_i} ; let i be minimal with this property. Then u has infinitely many opposite neighbours under π_{n_i} in G_{n_i} , and hence in $G_{n_i} - G_{n_{i-1}}$. Since π_{n_i} agrees with π on both U and $G_{n_i} - G_{n_{i-1}}$, our vertex u has infinitely many opposite neighbours also under π . Vertices in U_2 , finally, are happy with every π_{n_i} . By our choice of n_i , at least one of their opposite neighbours under π_{n_i} must lie in $G_{n_i} - G_{n_{i-1}}$. Since π_{n_i} agrees with π on both U_2 and $G_{n_i} - G_{n_{i-1}}$, this gives every $u \in U_2$ at least one opposite neighbour under π in every $G_{n_i} - G_{n_{i-1}}$. Hence u has infinitely many opposite neighbours under π , which clearly makes it happy. $\square$

8.6 Graphs with ends: the complete picture

In this section we shall develop a deeper understanding of the global structure of infinite graphs, especially locally finite ones, that can be attained only by studying their ends. This structure is intrinsically topological, because topology best captures our intuition about convergence.⁹

Our first goal will be to make precise our intuitive idea that the ends of a graph are the ‘points at infinity’ to which its rays converge. To do so, we shall define a topological space $|G|$ associated with a graph $G = (V, E, \Omega)$ and its ends.¹⁰ By considering topological versions of paths, cycles and spanning trees in this space, we shall then be able to extend to infinite graphs some parts of finite graph theory that would not otherwise have infinite counterparts; see the notes for more examples. Thus, the ends of an infinite graph turn out to be more than a curious phenomenon: they form an integral part of the picture, without which it cannot be properly understood.

To build the space $|G|$ formally, we start with the set $V \cup \Omega$. For every edge $e = uv$ we add a set $\dot{e} = (u, v)$ of continuum many points, making these sets $\dot{e}$ disjoint from each other and from $V \cup \Omega$. We then choose for each e some fixed bijection between $\dot{e}$ and the real interval $(0, 1)$, and

V, E, Ω

(u, v)

⁸ It is only by tradition that such partitions are called ‘unfriendly’; our vertices love them.

⁹ Only point-set topology is needed for the text. See the exercises for more.

¹⁰ The notation of $|G|$ comes from topology and clashes with our notation for the order of G . But there is little danger of confusion, so we keep both.

$[u, v]$ extend this bijection to one between $[u, v] := \{u\} \cup \overset{\circ}{e} \cup \{v\}$ and $[0, 1]$. This bijection defines a metric on $[u, v]$; we call $[u, v]$ a *topological edge* with *inner points* $x \in \overset{\circ}{e}$. Given any $F \subseteq E$ we write $\overset{\circ}{F} := \bigcup \{ \overset{\circ}{e} \mid e \in F \}$. When we speak of a ‘graph’ $H \subseteq G$, we shall often also mean its corresponding point set $V(H) \cup \overset{\circ}{E}(H)$.

Having thus defined the point set of $|G|$, let us choose a basis of open sets to define its topology. For every edge uv , declare as open all subsets of (u, v) that correspond, by our fixed bijection between (u, v) and $(0, 1)$, to an open set in $(0, 1)$. For every vertex u and $\epsilon > 0$, declare as open the ‘open star around u of radius ϵ ’, that is, the set of all points on edges $[u, v]$ at distance less than ϵ from u , measured individually for each edge in its metric inherited from $[0, 1]$. Finally, for every end ω and every finite set $S \subseteq V$, there is a unique component $C(S, \omega)$ of $G - S$ that contains rays from ω . Let $\Omega(S, \omega) := \{ \omega' \in \Omega \mid C(S, \omega') = C(S, \omega) \}$. For every $\epsilon > 0$, write $\overset{\circ}{E}_\epsilon(S, \omega)$ for the set of all inner points of S - $C(S, \omega)$ edges at distance less than ϵ from their endpoint in $C(S, \omega)$. Then declare as open all sets of the form

$$\hat{C}_\epsilon(S, \omega) \qquad \hat{C}_\epsilon(S, \omega) := C(S, \omega) \cup \Omega(S, \omega) \cup \overset{\circ}{E}_\epsilon(S, \omega).$$

$|G|$ This completes the definition of $|G|$, whose open sets are the unions of the sets we explicitly chose as open above.

closure $\overline{X}$ The *closure* of a set $X \subseteq |G|$ will be denoted by $\overline{X}$. For example, $\overline{V} = V \cup \Omega$ (because every neighbourhood of an end contains a vertex), and the closure of a ray is obtained by adding its end. More generally, the closure of the set of teeth of a comb contains a unique end, the end of its spine. Conversely, if $U \subseteq V$ and $R \in \omega \in \Omega \cap \overline{U}$, there is a comb with spine R and teeth in U (Exercise 83). In particular, the closure of the subgraph $C(S, \omega)$ considered above is the set $C(S, \omega) \cup \Omega(S, \omega)$.

standard subspace The subspaces X of $|G|$ we shall be interested in are usually the closure of a subgraph H of G , i.e., of the form $X = \overline{U} \cup \overset{\circ}{D}$ for $H = (U, D)$. We write $V(X)$ for U and $E(X)$ for D , and call such subspaces *standard*.

$V(X), E(X)$ We also refer to such X as $\overline{H}$, or even as $\overline{D}$ if H has no isolated vertices, and then say that X is *spanned by* D . Note that the ends in X are always ends of G , not of H ; in particular, they need not have a ray in H .

By definition, $|G|$ is always Hausdorff; indeed one can show that it is normal. When G is connected and locally finite, then $|G|$ is compact.¹¹

Proposition 8.6.1. *If G is connected and locally finite, then $|G|$ is a compact Hausdorff space.*

(8.1.2) *Proof.* Let $\mathcal{O}$ be an open cover of $|G|$; we show that $\mathcal{O}$ has a finite subcover. Pick a vertex $v_0 \in G$, write D_n for the (finite) set of vertices at distance n from v_0 , and put $S_n := D_0 \cup \dots \cup D_{n-1}$. For every $v \in D_n$,

¹¹ Topologists call $|G|$ the *Freudenthal compactification* of G .

let $C(v)$ denote the component of $G - S_n$ containing v , and let $\hat{C}(v)$ be its closure together with all inner points of $C(v) - S_n$ edges. Then $G[S_n]$ and these $\hat{C}(v)$ together partition $|G|$. $\hat{C}(v)$

We wish to prove that, for some n , each of the sets $\hat{C}(v)$ with $v \in D_n$ is contained in some $O(v) \in \mathcal{O}$. For then we can take a finite subcover of $\mathcal{O}$ for $G[S_n]$ (which is compact, being a finite union of edges and vertices), and add to it these finitely many sets $O(v)$ to obtain the desired finite subcover for $|G|$.

Suppose there is no such n . Then for each n the set V_n of vertices $v \in D_n$ such that no set from $\mathcal{O}$ contains $\hat{C}(v)$ is non-empty. Moreover, for every neighbour $u \in D_{n-1}$ of $v \in V_n$ we have $C(v) \subseteq C(u)$ because $S_{n-1} \subseteq S_n$, and hence $u \in V_{n-1}$; let $f(v)$ be such a vertex u . By the infinity lemma (8.1.2) there is a ray $R = v_0 v_1 \dots$ with $v_n \in V_n$ for all n . Let ω be its end, and let $O \in \mathcal{O}$ contain ω . Since O is open, it contains a basic open neighbourhood of ω : there exist a finite set $S \subseteq V$ and $\epsilon > 0$ such that $\hat{C}_\epsilon(S, \omega) \subseteq O$. Now choose n large enough that S_n contains S and all its neighbours. Then $C(v_n)$ lies inside a component of $G - S$. As $C(v_n)$ contains the ray $v_n R \in \omega$, this component must be $C(S, \omega)$. Thus

$$\hat{C}(v_n) \subseteq \hat{C}_\epsilon(S, \omega) \subseteq O \in \mathcal{O},$$

contradicting the fact that $v_n \in V_n$. □

If G has a vertex of infinite degree then $|G|$ cannot be compact. (Why not?) But $\Omega \subseteq |G|$ can be compact; see Exercise 91 for when it is.

What else can we say about the space $|G|$ in general? For example, is it metrizable? Using a normal spanning tree T of G , it is indeed not difficult to define a metric on $|G|$ that induces its topology. But not every connected graph has a normal spanning tree, and it is not easy to determine in graph-theoretical terms which graphs do. Surprisingly, though, it is possible to deduce the existence of a normal spanning tree from that of a defining metric on $|G|$. Thus whenever $|G|$ is metrizable, a metric can be made visible in a natural and structural way.

Theorem 8.6.2. (Diestel 2006; Pitz 2020)

For a connected graph G , the following assertions are equivalent:

- (i) *The space $|G|$ is metrizable.*
- (ii) *G has a normal spanning tree.*
- (iii) *All minors of G have countable colouring number.*

The proof of the equivalence of (i) and (ii) in Theorem 8.6.2 is indicated in Exercises 43 and 92. More on (iii) can be found in the notes.

Our next aim is to review, or newly define, some topological notions of paths and connectedness, of cycles, and of spanning trees. By substituting these topological notions with respect to $|G|$ for the corresponding graph-theoretical notions with respect to G one can extend to locally finite infinite graphs a number of theorems about paths, cycles and spanning trees in finite graphs whose ordinary infinite versions are false. We shall do this, as a case in point, for the tree packing theorem of Nash-Williams and Tutte, Theorem 2.4.1; see the notes for more.

X
connected Let X be an arbitrary Hausdorff space. (Later, this will be a subspace of $|G|$.) X is (*topologically*) *connected* if it is not a union of two disjoint non-empty open subsets.¹² Note that continuous images of connected spaces are connected. For example, since the real interval $[0, 1]$ is connected,¹³ so are its continuous images in X .

arc A *homeomorphic* image of $[0, 1]$ in X is an *arc* in X ; it *links* the images of 0 and 1, which are its *endpoints*. Every finite path in G defines an arc in $|G|$ in an obvious way. Similarly, every ray defines an arc linking its starting vertex to its end, and a double ray in G forms an arc with the two ends of its tails if these ends are distinct.

end degrees
in subspaces The (*topological*) *degree* of an end ω of G in a standard subspace X of $|G|$ is the supremum, in fact maximum, of all integers k such that X contains k arcs that end in ω and are otherwise disjoint.

$G; V, E, \Omega$ For the remainder of this section let, unless otherwise mentioned, $G = (V, E, \Omega)$ be a fixed connected locally finite graph.

Unlike ordinary paths, arcs in $|G|$ can jump across a cut without containing an edge from it – but only if the cut is infinite:

[8.7.1] **Lemma 8.6.3.** (Jumping Arc Lemma)
Let $F \subseteq E$ be a cut of G with sides V_1, V_2 .

- (i) If F is finite, then $\overline{V_1} \cap \overline{V_2} = \emptyset$, and there is no arc in $|G| \setminus \overset{\circ}{F}$ with one endpoint in V_1 and the other in V_2 .
- (ii) If F is infinite, then $\overline{V_1} \cap \overline{V_2} \neq \emptyset$, and there will be such an arc if both V_1 and V_2 are connected in G .

(8.2.2) *Proof.* (i) Suppose that F is finite. Let S be the set of vertices incident with edges in F . Then S is finite and separates V_1 from V_2 , so for every $\omega \in \Omega$ the connected graph $C(S, \omega)$ misses either V_1 or V_2 . But then so does every basic open set of the form $\hat{C}_\epsilon(S, \omega)$. Therefore no end ω lies in the closure of both V_1 and V_2 .

As $|G| \setminus \overset{\circ}{F} = \overline{G[V_1]} \cup \overline{G[V_2]}$ and this union is disjoint, no connected subset of $|G| \setminus \overset{\circ}{F}$ can meet both V_1 and V_2 . Since arcs are continuous images of $[0, 1]$ and hence connected, there is no V_1 – V_2 arc in $|G| \setminus \overset{\circ}{F}$.

¹² These subsets would be complements of each other, and hence also be closed. Note that ‘open’ and ‘closed’ means open and closed *in* X : when X is a subspace of $|G|$ with the subspace topology, the two sets need not be open or closed in $|G|$.

¹³ This takes a few lines to prove – can you prove it?

(ii) Suppose now that F is infinite. Since G is locally finite, the set U of endvertices of F in V_1 is also infinite. By the star-comb lemma (8.2.2), there is a comb in G with teeth in U ; let ω be the end of its spine. Then every basic open neighbourhood $\hat{C}_\epsilon(S, \omega)$ of ω meets $U \subseteq V_1$ infinitely and hence also meets V_2 , giving $\omega \in \overline{V_1} \cap \overline{V_2}$.

To obtain a V_1 – V_2 arc in $|G| \setminus \overset{\circ}{F}$, all we need now is an arc in $\overline{G[V_1]}$ and another in $\overline{G[V_2]}$, both ending in ω . Such arcs exist if the graphs $G[V_i]$ are connected: we can then pick a sequence of vertices in V_i converging to ω , and apply the star-comb lemma in $G[V_i]$ to obtain a comb whose spine is a ray in $G[V_i]$ converging to ω . Concatenating these two rays yields the desired jumping arc. $\square$

To some extent, arcs in $|G|$ assume the role that paths play in finite graphs. So arcs are important – but how do we find them? It is not always possible to construct arcs as explicitly as in the proof of Lemma 8.6.3 (ii). Figure 8.6.1, for example, shows an arc that goes through continuum many ends; such arcs cannot be constructed greedily by following a ray into its end and emerging from that end on another ray, etc.

There are two basic methods to obtain an arc between two given points, say two vertices x and y . One is to use compactness to obtain, as a limit of finite x – y paths, a *topological* x – y path, a continuous map $\pi: [0, 1] \rightarrow |G|$ sending 0 to x and 1 to y . A lemma from general topology then tells us that this path can be made injective:

Lemma 8.6.4. *The image of a topological x – y path in a Hausdorff space contains an x – y arc.* [8.7.3]

To illustrate this method, we will use it in the proof of Theorem 8.7.3.

Another method is to prove that the subspace in which we wish to find our x – y arc is topologically connected, and use this to deduce that it contains the desired arc. Our next three lemmas provide the tools needed to implement this approach in practice; we shall then illustrate its use in the proof of Theorem 8.6.9.

Being linked by an arc is an equivalence relation on the points of our Hausdorff space X : every x – y arc A has a first point p on any y – z arc A' (because A' is closed), and the obvious segments Ap and pA' together form an x – z arc in X . The corresponding equivalence classes are the *arc-components* of X . If X has only one arc-component, then X is *arc-connected*.

arc-
component

arc-
connected

Since $[0, 1]$ is connected, arc-connectedness implies connectedness. The converse implication is false in general, even for spaces $X \subseteq |G|$ with G locally finite. But it holds in all the cases that matter:

Lemma 8.6.5. *Connected standard subspaces of $|G|$ are arc-connected.* [8.7]

Our proof of Theorem 8.7.3 will show how one can prove Lemma 8.6.5. Two further proofs are indicated in Exercises 94 and 134.

[8.7]

Lemma 8.6.6. *Arc-components of standard subspaces of $|G|$ are closed.*

Proof. Let A be an arc-component of a standard subspace of $|G|$. Since A is connected, so is its closure $\overline{A}$. If $\overline{A} \setminus A \neq \emptyset$ then its points are limits of vertices in A (why?), so $\overline{A}$ is again standard. Hence $\overline{A}$ is arc-connected, either because $\overline{A} = A$ or by Lemma 8.6.5. But then $\overline{A} = A$, by definition of $\overline{A}$. Hence A is closed, as claimed. $\square$

Connected standard subspaces of $|G|$ containing two given points are much easier to construct than an arc between two points. This has to do with the fact that they can be described in purely graph-theoretical terms, with reference only to finite subgraphs of G rather than to $|G|$. The description can be viewed as a topological analogue of the fact that a subgraph H of G is connected if and only if it contains an edge from every cut of G that separates two of its vertices:

[8.7.1]

Lemma 8.6.7. *A standard subspace of $|G|$ is connected if and only if it contains an edge from every finite cut of G of which it meets both sides.*

Proof. Let $X \subseteq |G|$ be a standard subspace. For the forward implication, suppose that G has a finite cut $F = E(V_1, V_2)$ such that X meets both V_1 and V_2 but has no edge in F . Then

$$X \subseteq |G| \setminus \overset{\circ}{F} = \overline{G[V_1]} \cup \overline{G[V_2]},$$

and this union is disjoint by Lemma 8.6.3 (i). The induced partition of X into non-empty closed subsets of X shows that X is not connected.

The backward implication holds vacuously if X meets more than one component of G ; we may therefore assume that G is connected. If X is not connected, we can partition it into disjoint non-empty open subsets O_1 and O_2 . As X is standard, $U_i := O_i \cap V(X) \neq \emptyset$ for both i . Let $\mathcal{P}$ be a maximal set of edge-disjoint U_1 - U_2 paths in G , and put

$$F := \bigcup \{ E(P) \mid P \in \mathcal{P} \}.$$

Then $E(X) \cap F = \emptyset$, and no component of $G - F$ meets both U_1 and U_2 . Extending $\{U_1, U_2\}$ to a partition of V in such a way that each component of $G - F$ has all its vertices in one class, we obtain a cut $F' \subseteq F$ of G of which X meets both sides. As $E(X) \cap F = \emptyset$, it thus suffices to show that F is finite.

If F is infinite, then so is $\mathcal{P}$. As G is locally finite, the vertices of each $P \in \mathcal{P}$ are incident with only finitely many edges of G . We can thus inductively find an infinite subset of $\mathcal{P}$ consisting of paths that are

not only edge-disjoint but disjoint. As G is connected, the endvertices in U_1 of these paths have a limit point ω in $|G|$ (Proposition 8.6.1), which is also a limit point of their endvertices in U_2 . Since both O_1 and O_2 are closed in $|G|$, we thus have $\omega \in O_1 \cap O_2$, contradicting the choice of the O_i . $\square$

A *circle* in a topological space is a homeomorphic image of the unit circle $S^1 \subseteq \mathbb{R}^2$. For example, if G is the 2-way infinite ladder shown in Figure 8.1.3, and we delete all its rungs (the vertical edges), what remains is a disjoint union of two double rays; its closure in $|G|$, obtained by adding the two ends of G , is a circle. Similarly, the double ray ‘round the outside’ of the 1-way ladder forms a circle together with the unique end of that ladder.

circle

It is not hard to show that no arc in $|G|$ can consist entirely of ends. This implies that every circle in $|G|$ is a standard subspace; the set of edges spanning it will be called its *circuit*.

circuit

A more adventurous example of a circle is shown in Figure 8.6.1. Suppose G is the graph obtained from the binary tree T_2 by joining for every finite 0–1 sequence ℓ the vertices $\ell 01$ and $\ell 10$ by a new edge e_ℓ . Together with all the (uncountably many) ends of G , the double rays $D_\ell \ni e_\ell$ shown in the figure form an arc A in $|G|$, whose union with the bottom double ray D is a circle in $|G|$ (Exercise 100). Note that no two of the double rays in A are consecutive: between any two there lies a third (cf. Exercise 101).

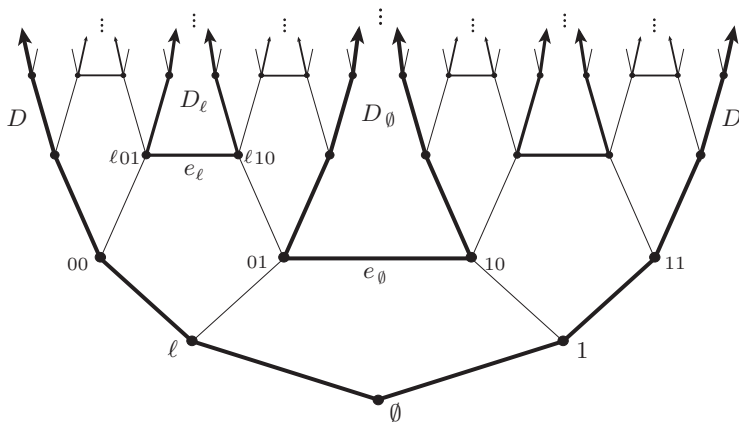


Fig. 8.6.1. The Wild Circle

A *topological spanning tree* of G is a connected standard subspace T of $|G|$ that contains every vertex but contains no circle. Since standard subspaces are closed, T also contains every end, and by Lemma 8.6.5 it

topological
spanning
tree

is even arc-connected. With respect to the deletion or addition of edges, it is both minimally connected and maximally ‘acirclic’ (Exercise 105).

One might expect that the closure $\overline{T}$ of an ordinary spanning tree T of G is always a topological spanning tree of $|G|$, but this is not the case: $\overline{T}$ may well contain a circle (Figure 8.6.2). Conversely, a subgraph whose closure is a topological spanning tree may well be disconnected: the ‘vertical’ rays in the $\mathbb{N} \times \mathbb{N}$ grid, for example, form a topological spanning tree together with the unique end.

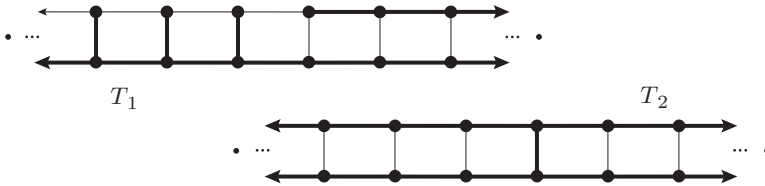


Fig. 8.6.2. T_1 is a topological spanning tree, but T_2 contains three circles.

Topological spanning trees can be constructed much as spanning trees of finite graphs: Lemma 8.6.11 will find one by iteratively deleting edges from $|G|$, but they can also be built up ‘from below’ (Exercise 108). Their mere existence even comes as a corollary of Theorem 8.2.4:

[8.7] **Lemma 8.6.8.** *The closure in $|G|$ of any normal spanning tree of G is a topological spanning tree of G .*

(1.5.4)
(8.2.3) *Proof.* Let T be a normal spanning tree of G . By Lemma 8.2.3, every end ω of G contains a normal ray R of T . Then $R \cup \{\omega\}$ is an arc linking ω to the root of T , so $\overline{T}$ is arc-connected.

It remains to check that $\overline{T}$ contains no circle. Suppose it does, and let A be the u - v arc obtained from that circle by deleting the inner points of an edge $f = uv$ it contains. Clearly, $f \in T$. Assume that $u < v$ in the tree-order of T , let T_u and T_v denote the components of $T - f$ containing u and v , and notice that $V(T_v)$ is the up-closure $[v]$ of v in T .

Now let $S := [u]$. By Lemma 1.5.4 (ii), $[v]$ is the vertex set of a component C of $G - S$. Thus, $V(C) = V(T_v)$ and $V(G - C) = V(T_u)$, so the set $E(C, S)$ of edges between these sets meets $E(T)$ precisely in f . Thus, $\overline{C}$ and $\overline{G - C}$ partition $|G| \setminus E(C, S) \supseteq A$ into two open sets both meeting A . This contradicts the fact that A is topologically connected. $\square$

Note that the proof of Lemma 8.6.8 did not use our assumption that G is locally finite: whenever a graph G has a normal spanning tree T , the closure of T in $|G|$ is an arc-connected subspace that contains no circle.